

Fraud detection in healthcare claims using machine learning: A systematic review

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ABSTRACT

Objective: Identifying fraud in healthcare programs is crucial, as an estimated 3%–10% of the total healthcare expenditures are lost to fraudulent activities. This study presents a systematic literature review of machine learning techniques applied to fraud detection in health insurance claims. We aim to analyze the data and methodologies documented in the literature over the past two decades, providing insights into research challenges and opportunities.

Methods: We identified research studies on health insurance fraud detection using machine learning approaches from databases such as Google Scholar, Springer-Link journals, Elsevier, PubMed, Excerpta Medica Database (EMBASE), Scopus, the Association for Computing Machinery (ACM) Digital Library, and the Institute of Electrical and Electronics Engineers (IEEE) Xplore Digital Library. We included only articles that presented experimental results of machine learning-based approaches applied to healthcare claims. From the reviewed articles, 137 were selected for the final qualitative and quantitative analyses.

Results: In recent years, there has been a surge in publications centered on the use of machine learning to detect health insurance fraud. Among these studies, those focused on the detection of fraud committed by healthcare providers was the most prevalent, followed by fraud committed by patients. A wide variety of machine learning algorithms are highlighted in these studies, ranging from unsupervised (41 studies) and supervised methods (94 studies), to hybrid approaches (12 studies). While traditional machine learning approaches remain dominant in this research area, the adoption of advanced deep learning techniques is on the rise. Considering the type of healthcare claims data used, 30 studies utilized private data sources, while the rest used publicly available datasets. Data from 16 countries were utilized, with a majority coming from the United States (96 studies), followed by China (11 studies) and Australia (5 studies).

Discussion and Conclusion: Detecting fraud in healthcare claims using machine learning presents several challenges. These include inconsistent data, absence of data standardization and integration, privacy concerns, and a limited number of labeled fraudulent cases to train models on. Future work should focus on enhancing transparency in data preparation, promoting the sharing of fraud investigation outcomes by authorities, and developing benchmark datasets to enhance accessibility and comparability. Furthermore, innovative techniques in data sampling, feature encoding methods for training machine learning models, and exploring the latest advancements in deep learning can significantly advance research in health insurance fraud detection.

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1. Introduction

Fraud, waste and abuse (FWA) in healthcare claims cost insurers, taxpayers, and those insured billions of dollars while putting beneficiaries' financial health and welfare at risk. The increasing enrollment in private and government healthcare programs, coupled with the rising costs and growing complexity of the healthcare system, heightens the risk and attractiveness of fraudulent activities [1,2]. In 2021, US healthcare spending rose by 9.7% from the previous year, reaching over 4.1 trillion USD, accounting for 19.7% of the Gross Domestic Product (GDP). [3]. Comparing across countries, for 2020, the healthcare spending in Australia was 202 billion AUD or 10.6% of GDP [4], while that in the UK was 257.6 billion GBP or 12% of GDP [5]. The National Health Care Anti-Fraud Association (NCHAA) of the US estimates that health insurance fraud cost the nation approximately 3%–10% annually, equating to approximately 120–410 billion USD for the year of 2021 [6]. According to an Australian health regulation expert, at least 7 billion AUD or 30% of the Australian Medicare scheme's funds are lost to Medicare fraud and billing errors per year [7].

The intricate nature of healthcare claims data poses significant challenges to the analysis of fraud detection. This data often exhibits high dimensionality and scale, lacks standardization and integration, and is characterized by multi-modality, heterogeneity, and non-stationarity. Compounding these challenges is the limited availability of ground-truth labels for fraud, which remains a major obstacle in implementing and validating machine learning (ML) models for the detection of fraudulent activities.

Recent advance in data mining and ML techniques, coupled with enhanced healthcare claims data collection methods and availability of data repositories, present a promising opportunity to improve fraud detection systems through data analytics. This paper explores the potential of using ML for health insurance fraud detection, in the context of recent literature. Our main contributions include a summary of available literature on this topic, a discussion of the challenges in detecting such fraud, and recommendations for future research in healthcare claims fraud detection. These insights are drawn from a review of 137 research papers focused on fraud detection in healthcare claims using ML over the past two decades.

More specifically, we aim to address the following research questions:

1. *Types of fraud*: Which individuals or entities are responsible for committing healthcare claims fraud, and what are the prominent types of fraud discussed in the literature?
2. *Datasets*: What are the available datasets for detecting healthcare claims fraud?

3. *Methods*: What processes and transformations are applied to healthcare claims data for analytical purposes? Which ML methods and algorithms are commonly used in fraud detection from healthcare claims, and how effective are these methods?

4. *Challenges*: What are the current challenges in detecting fraud from healthcare claims?

5. *Opportunities*: What areas present potential for further exploration in healthcare claims fraud detection?

In this study we aim to provide valuable insights to researchers and practitioners in this domain about the techniques and algorithms that have been developed, implemented and evaluated in literature. We also suggest additional avenues which can be explored within the ML space that could enhance fraud detection methodologies from healthcare claims data. Additionally, healthcare claims fraud investigators and practitioners may benefit from our insights, as we highlight applications based on real-world data and their efficacy in identifying fraud from healthcare claims.

2. Background

The healthcare and health insurance systems are complex, encompassing a multitude of actors, variables, and processes. Within these systems, primary actors include patients, healthcare providers, and insurers. Generally, when a patient visits a provider, a claim is filed (by either the provider or the patient) and forwarded to insurers for payment of the services rendered. When both provider and patient are enrolled in health insurance programs, the provider submits the claim directly to the insurer. However, if the provider is not registered with such a program, they invoice the patient, who pays the bill. The patient then pays this bill out-of-pocket and subsequently files a claim for either full or partial reimbursement [8]. While various forms of fraud may occur within this system, for example overbilling and unbundling, it is important first to define what constitutes health insurance fraud.

The Center for Medicare and Medicaid Services (CMS) defines FWA in health insurance programs as follows [9]:

1. Knowingly submitting, or causing to be submitted, false statements or creating misrepresentations of fact to obtain a payment or claims reimbursement for which no entitlement would otherwise exist;
2. Knowingly soliciting, receiving, offering, and/or paying remuneration to induce or reward referrals for items or services reimbursed by federal healthcare programs or private healthcare insurers. This includes kickbacks, bribes, or rebates;
3. Making prohibited referrals for certain designated health services;

- 4. Exhibiting practices that result, either directly or indirectly, in unnecessary costs and it includes any practice that does not provide patients with medically necessary services or meet professionally recognized standards of care.

In the health insurance system, a multitude of variables are susceptible to exploitation by various actors, which complicates the detection of fraud. Providers, for instance, can manipulate elements like pricing, medical procedures, drug prescriptions, specialty services, and the complexities of diverse services to their advantage. Examples of such fraud include upcoding (claims with reimbursement values that exceed the actual value of the services provided) [2,10–16] and kickbacks (collusion between provider and patient to gain commissions for unauthorized services) [17–19].

3. Materials and methods

Following standard systematic literature review practices, our paper selection was based on criteria specifically tailored to this review’s themes and research objectives. Moher et al. [20] have established guidelines for reporting meta-analyses and scientific reviews, known as PRISMA (Preferred Reporting Items for Systematic reviews and Meta-Analyses), and this survey paper follows its specifications as well as the systematic review paper layout as outlined by Si et al. [21].

Papers eligible for inclusion are characterized by (a) a focus on fraud detection, (b) the utilization of healthcare claims data, and (c) the application of ML algorithms and models. This review focuses solely on studies where ML is the primary method for detecting fraud. This specific emphasis is due to the unique attributes of healthcare claims data, which render ML techniques more suitable than other data-driven analytical methods, as detailed in the Introduction. Future progress in detecting healthcare claims fraud is expected to be predominantly influenced by advancements in ML and deep learning (DL) techniques. Guided by the aforementioned inclusion criteria, we used the following query in our literature search:

“healthcare” OR “health insurance” OR “medicare” OR “medicaid”) AND (“fraud” OR “waste” OR “abuse” OR “FWA”) AND (“machine learning” OR “data science” OR “neural networks” OR “deep learning”)

This query was executed on 30 July 2024 against the following online academic literature databases: Google Scholar, Springer-Link journals, Elsevier, PubMed, EMBASE, Scopus, the Association for Computing Machinery (ACM) Digital Library, and the Institute of Electrical and Electronics Engineers (IEEE) Xplore Digital Library. The query was iteratively refined by testing various combinations of keywords. The goal was to cast a wider net to capture as many relevant articles as possible. However, we acknowledge that despite this recall-oriented approach, some articles might not have been identified in our search, a limitation common to many systematic reviews [22–24].

The database search targeted peer-reviewed journals and conference proceedings published from 2000 to 2024. Patents, research theses, proceeding summaries, posters and pre-prints were excluded. Furthermore, studies that used healthcare claims data for purposes other than fraud detection, did not offer experimental results, or presented decision support systems for fraud detection without real-world data applications, were excluded. To summarize, we implemented the inclusion and exclusion criteria detailed in Table 1 to select studies that are potentially relevant for addressing our research questions, as stated in the Introduction.

After retrieving relevant articles from the databases and conducting full-text screening, each article was thoroughly reviewed to ensure its relevancy. During this process, the criteria stipulated in Table 1 were strictly followed.

Since the primary objective of this paper is to discuss practical applications of fraud detection from healthcare claims, papers which did not cover both healthcare fraud and ML applications, nor mentioned experimental data in their abstracts were omitted. Additionally,

Table 1
The inclusion and exclusion criteria.

Inclusion criteria	Exclusion criteria
Articles published from the year 2000 to mid-2024.	Any older research papers were omitted as they are deemed outdated.
Papers which are characterized by (a) a focus on fraud detection, (b) the use of healthcare claims data, and (c) the application of ML algorithms and models.	Papers utilizing healthcare claims data and ML for purposes other than fraud detection.
Research papers published in peer-reviewed journals and conference proceedings.	Patents, research theses, proceeding summaries, posters and pre-prints.
Papers which utilized real-world data and report in detail their application purpose, data processing, the ML techniques employed and performance results, as per the research questions stated in the Introduction.	Papers which used hypothetical or simulated data, incomplete from a data processing perspective, irrelevant to fraud detection from healthcare claims, duplicates, solely based on statistical analysis, and theoretical in nature.

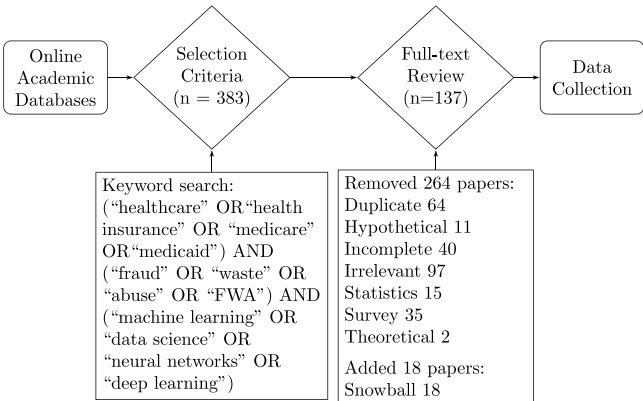


Fig. 1. The methodological framework for this survey paper.

as there is a focus on only journal and conference articles, the initial search results were reduced to a set of 383 articles.

Next, each article was thoroughly reviewed to ensure its applicability. Articles utilizing simulated or hypothetical data were also excluded due to their lack of real-world applicability. Papers were deemed incomplete when they omitted details, such as, dataset country of origin, dataset size, label distribution, features used, and dataset splits for train, test and/or validation in supervised learning ratios (where applicable to the applications). The omission of these crucial details may render such research articles unreliable as a reference point for understanding the application’s objectives, and comparing outcomes with related literature, since they reduce research reproducibility and comparison capabilities. This is to ensure conformity with research questions 2 and 3. Several papers, while focusing on fraud detection from healthcare claims, utilized non-ML systems engineering approaches and were thus deemed irrelevant as they are outside the scope of our review. Articles that solely employed statistical methods were omitted since this review focuses on ML applications. Duplicate publications were removed as well. An additional 53 papers were identified using a snowballing strategy, wherein references from survey or review papers were traced, however it was found that 35 papers were duplicates. Hence 18 new and relevant articles were discovered through snowballing.

This narrowed the 383 identified publications to the final set of 137 relevant articles. Fig. 1 presents a condensed PRISMA diagram outlining our literature selection methodology, which directed the gathering and selection of research articles for discussion in this survey paper.

In order to structure our analyses, we extract the following data points from the final set of 137 included papers:

- Publication details: the conference or journal where the paper was published, and the date of publication.
- Types of fraud: the actor(s) under investigation (i.e. provider, insurer, or patient), and the specific types of fraud under study.
- Dataset details: the dataset(s) used, their source, and the countries of origin. The dataset size, number of features and the data preprocessing activities were also noted. For supervised learning approaches, the label distributions and as well as the train, test, and/or validation ratios were documented.
- Modeling objectives: the objective of the ML algorithms i.e. the desired purpose of fraud prediction models.
- ML approaches: the types of learning, ML approaches, and specific ML algorithms employed.
- Evaluation metrics: quantitative measurements of performance for the evaluation of ML approaches using suitable metrics.

4. Other survey papers

Several survey papers were identified during our literature search process. For example, Bauder et al. [2] reviewed fraud detection in healthcare claims but focused primarily on healthcare upcoding fraud. Their study reviewed 26 papers, 11 of which were upcoding application-based papers from 2000 to 2014, while the remaining 15 were survey papers post-2010. However, they did not perform a systematic review, and the article lacked details on research methodology, databases used, and search criteria. Upcoding fraud is only one type of health insurance fraud, and hence this study covers only a subset of articles we intend to discuss in our survey.

Ngai et al. [25] performed a systematic literature review on the application of data mining techniques for financial fraud detection across different sectors including banking, insurance, securities and commodities. Their category of insurance fraud encompasses crop, healthcare and automobile insurance fraud. The literature review covering publications from 1997 to 2008 divided data mining techniques into six classes: classification, clustering, outlier detection, prediction, regression and visualization. Out of 49 articles reviewed, only five pertain to fraud in health insurance. These five articles are also included in our review. Unlike the aforementioned review articles, this paper provided details on research methodologies, databases utilized, and specific keyword searches and corresponding results. However, the broad scope of their review diverges from the more focused scope of our study.

Li et al. [26] summarized key findings from approximately 40 research articles aimed at detecting health insurance fraud using statistical methods, published between 1993 and 2008. They provided high-level summaries of the literature and emphasized the application of statistical analysis (SA) and ML to data, instead of discussing the applications in-depth. They did not conduct a systematic review, provide their research methodology, discuss the databases employed, or disclose their search criteria. This work diverges from the objective of our study, which is centered on presenting exclusively ML applications with thorough analysis. Additionally, the range publication years covered is outdated.

Joudaki et al. [27] investigated 21 data mining application papers from 1997 until 2013 targeting health insurance fraud. The ML approaches utilized by applications studied in this paper include SA, clustering, outlier detection (OD), association rule mining (ARM), decision trees (DTs), neural networks (NNs), support vector machines (SVMs), and genetic algorithm (GA). An emphasis was placed on applications using “original data”, though details of such data is not clearly explained. Their review lacks specific details on research methodologies, databases used, and their search criteria. While their work serves as a suitable introduction to research on fraud detection from healthcare claims, it is outdated.

Abdallah et al. [28] discussed challenges that negatively impact the performance of fraud prevention systems in five electronic commerce

systems, namely credit card, telecommunication, healthcare insurance, auto-mobile insurance, and online auction. Their discussion spanned publications from 1994 to 2014, with fourteen being applications in fraud detection from healthcare claims. These publications are also reviewed in our study. While Abdallah et al. focused on fraud detection systems and discuss their mechanisms, they failed to specify their research methodology, databases used, or keywords used for their searches. Their work differs from the emphasis of this survey paper as it centers on general fraud detection systems without stressing real-world data-driven applications. Moreover, the range of publication years covered by their study is outdated.

Albsharawi [29] examined data mining applications for financial fraud detection, focusing on four major categories: financial statement fraud, bank fraud, insurance fraud, and other related financial fraud. Bank fraud was further segmented into credit card fraud, money laundering, and fraudulent bank accounting. Insurance fraud was segmented into healthcare, automobile, and corporation frauds. In total, they reviewed 65 articles published from 2004 to 2015, three of which are applications in fraud detection from healthcare claims. These three articles are also reviewed in our paper. The authors discuss the databases and keywords used for their searches, namely “detecting financial fraud”, “financial fraud and data mining”, “financial fraud detection”, and “detecting financial fraud via data mining”. Similar to Ngai et al. [25], the scope of this paper is too broad compared to the focus of our study.

Sheshasaayee and Thomas [30] exclusively focused on supervised learning methods, a subset of ML approaches, applied in health insurance fraud detection. They reviewed 10 research papers published from 2002 to 2016. However, they did not conduct a systematic review, provide their research methodology, discuss the databases utilized, or provide their search keywords. In contrast to our study, this article solely covers supervised learning applications for fraud detection from healthcare claims.

Bauder and Khoshgoftaar [31] discuss Medicare fraud analysis and detection research, with an emphasis on data processing and integration. They reviewed approximately 25 papers published between 2009 and 2018. However, they do not disclose their research methodology, databases utilized, or search keywords. Although they briefly discussed ML applications in fraud detection from healthcare claims, their primary focus was on challenges and opportunities in the current research on Medicare data processing and integration, contrasting with the objective of this survey paper.

Ai et al. [32] performs a systematic review on healthcare fraud detection. They reviewed 27 papers published between 2001 and 2016. They disclose their research methodology, keywords, and databases utilized, however their search does not focus on ML applications specifically and offers too broad a review. Furthermore, their evaluation of the papers do not offer detail such as healthcare data sources, data preparation, ML models, model performance, etc. In conclusion, the scope of this paper is too broad and the summaries offered to sparse compared to the focus of our study.

Other and more recent survey papers include surveys on financial fraud detection [33–35] or even broader fraud detection [36–40], which are too broad compared to the focus of our study. They, together with the healthcare fraud detection surveys [41–49] do not offer the detail we require.

In summary, our survey paper distinguishes itself from the aforementioned works in several ways. Firstly, this study covers more recently published works, spanning a wider time frame from 2000 to 2024. Secondly, our work reports on recent innovations and technical advancements in the field, as well as more relevant applications, specifically utilizing only real-world data. Our review excludes theoretical works, works with simulated data, or research papers that provide sparse details on the work performed (e.g., if the exact algorithm, dataset size, label distribution, and number of features used are not disclosed, the research paper is not included in this work). To the extent of our knowledge, this survey paper is the first to provide an in-depth review of dataset specifications and the ML algorithms applied to fraud detection from healthcare claims.

Table 2

The distribution of fraud applications by actor type.

Actor	Type	# Appli.	Reference
Conspir- acy	U	1	[50]
	S	4	[51–54]
Patient	U	6	[55–60]
	S	6	[19,61–65]
Provider	U	38	[8,10–16,18,50,58,66–92]
	S	84	[13,19,52,62,74,93–138,138–170]
Provider/ Insurer	S	6	[171–176]

5. Results and findings

The following section presents the insights gathered from the review process. Various aspects of health insurance fraud are discussed, encompassing the individuals and entities involved, relevant datasets and associated challenges, data sampling and preprocessing activities, and diverse learning approaches employed for fraud detection.

5.1. Fraud actors

The health insurance system experiences a wide range of fraud types, committed by individuals and groups, the latter referred to as “conspiracy fraud” in this paper. Table 2 shows how the health insurance fraud articles are distributed by actor type, and furthermore distinguishes between unsupervised and supervised learning settings.

‘Provider fraud’ is identified as the most common type in the literature. In supervised learning applications, such fraud is predominantly detected by identifying fraudulent claims for procedures, drugs, or visits associated with specific providers. For unsupervised learning applications, anomalies like mismatches between a provider’s specialty and performed procedures, or between diagnoses and treatments are investigated. Additionally, outliers in claim amounts or discrepancies in claims compared to similar providers in the same region were analyzed.

When addressing ‘patient fraud,’ healthcare data was examined for indicators of prescription shopping (often for controlled substances), doctor shopping, and other treatment misuse behaviors.

Research papers which explore erroneous payments made by insurers – attributable either to the insurer or provider committing fraud – are tagged as ‘provider/insurer fraud.’ In the case of conspiracy, two or sometimes all three actors collaborate in executing the fraud. While a majority of the reviewed articles concentrate on fraud involving a single specific actor, Kosea et al. [50] and Johnson et al. [52] both probe applications that might pertain to either provider or conspiracy fraud. Additionally, three research studies [19,58,62] examine both provider and patient fraud. Furthermore, Chandola et al. [13] discuss three applications relevant to providers, each distinguished by its own dataset. Hence, a total of 145 applications were investigated from the 137 papers.

5.2. Datasets

The persistent global challenge of detecting fraud in healthcare claims is underscored by several decades of research conducted worldwide. The breadth of this issue is further highlighted by the geographic diversity of data sources cited in the articles reviewed for this study, as illustrated in Table 3. Although the majority of publications rely on data from a single country, Naidoo et al. [113] is a notable exception, utilizing two distinct datasets—namely, public data from the US (CMS), and private insurance data from a South African (S. Africa) company. Despite the worldwide presence of both government-sponsored and private health insurers, research contributions often focus predominantly on specific countries, as illustrated in Table 3, in which the number of

Table 3

The data source distribution by country.

Country	#	Ave. Size	Ave. Ratio	Reference
US	96	13,179,589	99.9:0.1	[8,10–14,16,52,54,56,61,64,66,71,74,75,78,79,81–84,88,90–96,98,99,102–111,113–120,124,126–134,136–138,138–142,144–151,153–172,176]
China	11	3,759,969	86:14	[53,57,58,60,65,85,89,123,135,152,175]
Aus	5	8,078	NA	[19,55,68,69,73]
Germany	4	501,804	98.6:1.4	[59,143,173,174]
India	4	390,128	90:10	[62,112,122,125]
Iran	3	24,068,311	99.98:0.02	[77,86,87]
Taiwan	3	1,661	81.4:19.6	[80,100,101]
Turkey	3	580,460	NA	[50,67,70]
Chile	2	6,449	95:5	[18,51]
Ghana	1	530	40:60	[121]
Italy	1	396,246	NA	[15]
Korea	1	3,705	90:10	[97]
NL	1	40,000,000	NA	[72]
Nigeria	1	2,477	NA	[76]
S. Africa	1	3,567	90:10	[113]
S. Arabia	1	396	36:64	[63]

applications per country, average raw data set size before processing and average balance ratios after pre-processing are reported. In unsupervised learning settings, labels are not required and the balance ratios are not reported, hence ‘NA’ is noted for their balance ratios and the real-valued reported averaged balance ratios are those of the supervised applications.

This diversity in data sources is multifaceted: insurance claims data include details of services rendered and patient demographics; provider data encompass specialties, locations, and other relevant information about healthcare providers; and clinical-instance data comprise a range of clinical information on patients, such as histopathology and radiology reports [28].

Examples of public data sources attributable to the US include the US Healthcare Financing Administration (HCFA) [177], the Office of the Inspector General’s (OIG) List of Excluded Individuals and Entities (LEIE) [178], and CMS repositories [179], such as the CMS Medicare Part B Provider Utilization and Payment Data including Physician and Other Supplier data (PUP_POS) [180], Part D Prescribers (PUP_D) [181], and the Durable Medical Equipment, Prosthetics, Orthotics, and Supplies (DMEPOS) [182] datasets. Furthermore, the National Plan and Provider Enumeration System (NPPES) with its National Provider Identifiers (NPIs) and the Healthcare Common Procedure Coding System (HCPCS) [183] are utilized to link providers to the CMS datasets and decode procedure codes. Among the reviewed studies, 71 of the 137 (51%) papers utilized CMS data since they are reliable, updated periodically, publicly available, and can be used to derive labeled datasets. Although a universally accepted benchmark dataset for model performance comparison is lacking, the CMS data can serve as a suitable proxy, as the data is available without cost, has labels and offers large datasets for model training. Khoshgoftaar et al. are prominent researchers in the field of fraud detection from healthcare claims, with 41 of their publications reviewed in this paper. These publications all utilize CMS datasets. Their research on CMS dataset integration, feature selection, and preprocessing serves as a valuable starting point for preparing datasets for ML modeling [11,75,102,104,105,107,108].

As seen in Table 3 applications using USA datasets are most prominent in literature. The applications can be subdivided into CMS and

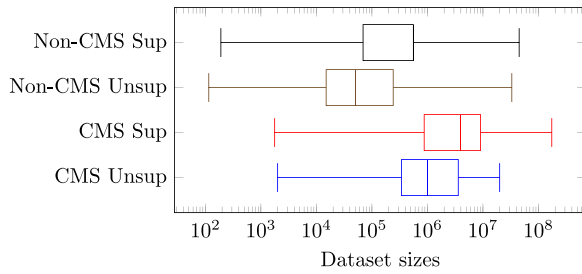


Fig. 2. The distribution of USA data set sizes for CMS and non-CMS datasets, for unsupervised and supervised applications.

non-CMS data, with 69 and 27 applications respectively. Fig. 2 provide boxplots for the dataset sizes for the different applications using USA data. In general, CMS datasets are much larger than non-CMS datasets and supervised applications enjoy larger datasets than unsupervised applications.

Australia's public data sources feature Health Insurance Commission (HIC) [184], Medicare Australia's Enterprise Data Warehouse, and Medicare Australia's (MA) repositories such as the Pharmaceutical Benefits Scheme data (PBS) and the Medicare Benefit Schedule data (MBS) [185].

Other sources of country-specific data include National Health Insurance (NHI) [80,100,101], National Health Insurance Corporation (NHIC) [97], Health Insurance Review and Assessment Services (HIRA) [97], Health Insurance Scheme (HIS) [76,121], Health Maintenance Organization (HMO) [76] and Health Insurance Management Center [57].

Private datasets, often sourced from private insurers, have limited descriptions due to privacy considerations. To obtain more detailed information about these datasets, it is necessary to reach out to the authors of the papers that utilized them [13,18,51,52,88,99,113,174].

We observe that healthcare claims, despite their diversity in content and richness across different sources and regions, often share similarities when integrated into ML workflows. For patient profile representation, features typically encompass aspects like patient gender, age, ethnicity, insurance provider, insurance type, diagnosis, and received treatment. For provider profile representation, features often include provider location, specialty, typical diagnoses observed, and typical procedures/services and their quantity/units performed. Meanwhile, features derived from procedure claims usually comprise of patient data, diagnosis codes, procedure codes, claim amounts, reimbursement amounts, number of distinct beneficiary/day services, number of distinct beneficiaries receiving the procedure/service, and associated provider data [95,180]. Features derived from drug prescription claims are usually comprised of the following: provider information, the beneficiaries receiving the drugs, the drugs the providers administered, the number of standardized 30-day fills, number of day's supplies and cost paid for all associated claims [95,181]. For claims related to medical equipments, features often include provider details, suppliers used by providers, the beneficiaries associated by the suppliers, claims submitted by a supplier from a referring order, services/products rendered by a supplier, claim amounts and reimbursement amounts [95,182].

In supervised ML applications, having data labeled as either fraudulent or legitimate is essential. These applications are typically framed as binary classification tasks. However, such labels are often scarce, and the datasets tend to be highly imbalanced—a topic extensively discussed in Sub Section 5.6.3. On average, dataset sizes stand at a few million instances, with the smallest being a few hundred instances [82] and the largest being around eighty million instances [124], with fraudulent cases typically comprising of 0.1 to 10% of overall instances.

Table 4

The distribution of sampling methods.

Sampling method	# Appli	Reference
ROS	10	[94,103,104,107,110,126,127,134,151,173]
ROS-RUS	6	[94,103,110,119,126,147]
RUS	31	[62,84,94,96,98,100,103,104,107–111,114,117,120,126,127,134,136,141,144–146,151,152,157,166,169,173,175]
SMOTE	16	[62,63,107,118,127,133,134,137,150,151,158,159,163,164,173,175]
SMOTE	16	[62,107,151,156,159,173]
Other	9	[62,107,127,134,151,156,159,175]

Applications of unsupervised methods primarily concentrate on identifying outliers within datasets. Such outliers may appear in various forms, including anomalies in claim amounts [15,76,79], discrepancies between a provider's specialty and the procedures they perform [77,78], and mismatches between diagnoses and corresponding treatments [132].

5.3. Data sampling methods

Diverse sampling methods are prevalent in the literature to tackle the wide-spread challenge of label scarcity and imbalance. Random over-sampling (ROS) and random under-sampling (RUS) are two widely accepted data sampling techniques.

ROS increases the number of the minority or positive (fraud) instances by duplicating existing positive instances at random. The disadvantages of this technique include increased dataset size which leads to increased computational demands, and an increased risk of overfitting by ML models due to duplicate instances.

Conversely, RUS reduces the number of majority or negative (non-fraud) instances by sampling from the negative instances without replacement and subsequently discarding them from the negative class. The reduction in dataset size leads to less demanding computational requirements, but at the cost of potentially discarding valuable data. Our review indicates a general preference for RUS over ROS as indicated in Table 5 and [96,98,108,109,136,141,144,145,157,169,186]. Some research works also employ a hybrid ROSRUS technique [94,103,110,147]. For detailed comparisons between different sampling methods, refer to Bauder and Khoshgoftaar [107] where they compare the performance of three different classifiers trained on five different class distributions, which were created by six different sampling methods, namely ROS, RUS, Synthetic Minority Oversampling Technique (SMOTE), Borderline-SMOTE (SMOTEb1 and SMOTEb2), Adaptive Synthetic SMOTE (ADASYN). Furthermore, majority weighted minority oversampling (MWMO or MWMOTE) and random walk oversampling (RWO) are notable variants of ROS [127] utilized by researchers. Other works include customized algorithms for sampling [151,156,159,164].

Of the 137 reviewed papers, 44 employ one or more sampling techniques and the relevant references are available in Table 4. Only two articles used sampling for an unsupervised learning setting [84,104] and the rest for supervised applications. Table 5 provides a summary of the AUC performances of frequently used supervised algorithms (identified in Table 8), trained and tested on balanced (50:50 ratio for non-fraud: fraud) CMS Part B dataset across frequently used and different sampling methods. AUC has been reported as it is the most frequently occurring metric - 69 of the 145 applications use AUC (47%), 42 use accuracy, 41 use F1-score and 28 use precision. The average performance is provided together with the minimum and maximum reported performance in brackets respectively. The maximum reported AUC per algorithm is set in bold text. For a clarification of algorithm abbreviations, please refer to Section 5.5. As noted before, a universally accepted benchmark dataset for healthcare fraud detection is not

Table 5

The AUC performance for different supervised ML algorithms on different sampling methods for both imbalanced and balanced distributions of CMS Part B datasets.

Ratio (non-fraud: fraud)	CatB (min, max)	DT (min, max)	GBT (min, max)	LightGBT (min, max)	LogR (min, max)	NN (min, max)	RF (min, max)	XGB (min, max)	Reference
Imbalance (99:95:0.05 ave)		0.50	0.79		0.82	0.81	0.73 (0.63, 0.79)		[107,108,110,134]
50:50 ROS			0.81		0.82	0.85	0.78 (0.75, 0.80)		[94,103,107,110,134]
50:50 RUS	0.86 (0.79, 0.91)	0.81 (0.77, 0.87)	0.77 (0.72, 0.81)	0.78	0.79 (0.65, 0.87)	0.78 (0.66, 0.86)	0.81 (0.76, 0.85)	0.78 (0.71, 0.86)	[94,96,98,103,107–110,114,117,134,136,145,146,157]
50:50 ROS-RUS						0.81 (0.70, 0.85)			[94,103,110,147]
50:50 SMOTE			0.73		0.82		0.79		[107,134]
50:50 ADASYN			0.69		0.82		0.74		[107]

recognized, hence CMS Part B data is used as proxy to facilitate model performance comparison in this paper as 50 of the 145 applications use CMS Part B (34%) and 24 use CMS Part D, while 9 use DMEPOS.

From Table 5 it is clear that RUS is the preferred method, as 15 studies employ it compared to the other sampling methods. RUS also consistently reports the highest performance values for the supervised algorithms. It is interesting to note that GBTs, LogR, NNs, and RFs perform only slightly worse on imbalanced datasets, which could indicate their applicability in severely imbalanced cases.

Table 6 provides a summary of the AUC performances of frequently used supervised algorithms trained and tested on CMS Part B data across frequently observed sampling distributions via the RUS method. The average performance is provided together with the minimum and maximum reported performance in brackets respectively. The maximum reported AUC per algorithm is set in bold text. Hancock et al. [141] reported unusually high AUC values for their study with RF and XGB, which is skewing the data for these algorithms. The RF algorithm proves to be most popular i.t.o. frequency of use, with seven applications for both the 99:1 and 90:10, five applications for the 75:25 and six applications for the balanced distributions. Thereafter LogR is most frequently used with five applications for both the 75:25 and 65:35 distributions, as well as seven applications for the balanced case.

The data is slightly skewed by the reported results of Hancock and Khoshgoftaar [117], where performances range from 0.50–0.52 for a ratio of 99:1, 0.55–0.62 for a ratio of 90:10 and 0.65–0.79 for a ratio of 50:50 for CatB, LightGBT, LogR and RF. As expected, the performance improves with more balanced datasets, however these are poor results, as a performance of 0.50 relates to random guessing, meaning the model has not learned the prediction task. Compared to the same distributions, data and models [98,107,108], the difference is the size of the datasets used, where [117] used 47 million instances and the others 1–2 million. This illustrates that data selection and per-processing activities are fundamental to the success of the learning.

5.4. Data preprocessing activities

Various methods can be employed to preprocess data before using them in train ML models. For example, semantic embeddings (SEs), dimensionality reduction algorithms, and other feature engineering techniques are popular preprocessing approaches.

Recent studies delve into innovative data preprocessing strategies. For instance, one-hot encoding of categorical variables with thousands of distinct values can produce high-dimensional vectors, increasing model complexity and resulting in large, sparse datasets.

Moreover, this method often fails to represent inherent relationships in the data. Semantic embeddings (SEs) tackle this issue by generating low-rank continuous vectors that encapsulate multiple dimensions of similarity [100,105,109,128]. A variety of semantic embedding methods exist, such as one-hot encoding (OHE), continuous-bag-of-words (CBOW), skipgram (SG), GloVe, Med-W2V (or Med-Word2Vec), HcpcsVec, Hcpcs2Vec, RxVec, and the aggregated method (AM) [111,116,119,128,139].

Table 7 provides a summary of the AUC performances for different SEs used with frequently used supervised ML algorithms, trained and tested on imbalanced CMS part B data with an average ratio of non-fraud to fraud of 99:1. The maximum values per algorithm are set in bold text. The average performance is provided together with the minimum and maximum reported performance in brackets respectively. The maximum reported AUC per algorithm is set in bold text.

Among the reviewed methods, Hcpcs2Vec leads to the best overall algorithm performance. As seen in Table 7, Hcpcs2Vec paired with either GBT and XGB demonstrates superior LEIE fraudulent provider classification performance with the highest AUC values of 0.87, compared to cases without SEs with an average AUC of 0.84 [116,128]. CBOW, GloVe and SG, when combined with GBT, achieve competitive results for LEIE fraudulent provider classification as well.

Dimensionality reduction techniques aim to reduce the number of feature dimensions in a dataset in the effort to reduce the memory requirements when training and deploying ML algorithms, and avoid the *curse of dimensionality* [187], among other advantages. Notable techniques include principal component analysis (PCA), discriminant analysis classifier (DAC, sometimes known as linear discriminant analysis) and quadratic discriminant analysis (QDA) [127]. PCA indicates the most significant features by capturing the relationship between the features and their contribution towards the variance in the data. It is often used as a pre-processing step to supervised learning [116,118,119,127,133] or as an unsupervised technique [74,80]. Conversely, DAC [74,127] and QDA [127] can be used as supervised learning techniques since they require labels as inputs. These dimensionality reduction techniques can be applied in combination with SE [105,109]. From the literature review it was noted that the use of PCA rarely improved model performance, even in scenarios with varied sampling methods [127,132,133].

An alternative approach involves utilizing feature selection methods, such as the Greedy Feature Selection Algorithm (FSA), the Laplacian Score (LS), the Spectral Algorithm (Spec), and the Unsupervised Spectral Feature Selection Method (USFSM) as recently explored by

Table 6

The AUC performance for different supervised ML algorithms on different RUS distributions of CMS Part B datasets.

Ratio (non-fraud: fraud)	CatB (min, max)	DT (min, max)	GBT (min, max)	LightGBT (min, max)	LogR (min, max)	NN (min, max)	RF (min, max)	XGB (min, max)	Refe-rence
99.95:0.05			0.79		0.82	0.97	0.76	0.97	[108,141]
99.9:0.1		0.80			0.81	0.81	0.74		[96,98,110]
99:1	0.51	0.80	0.78 (0.75, 0.80)	0.52	0.75 (0.50, 0.83)	0.76 (0.65, 0.81)	0.78 (0.50, 0.97)	0.74 (0.50, 0.97)	[96,98,103, 107,108, 110,117, 141,157]
95:5		0.79	0.78		0.78 (0.74, 0.82)	0.72	0.87 (0.77, 0.97)	0.97	[96,98,141, 157]
90:10	0.62	0.79	0.80 (0.77, 0.82)	0.622	0.75 (0.52, 0.83)	0.75 (0.68, 0.81)	0.80 (0.58, 0.97)	0.79 (0.61, 0.97)	[96,98,107, 108,110, 117,141]
80:20		0.88	0.72		0.82 (0.69, 0.88)	0.76 (0.66, 0.81)	0.73		[103,110, 136,145, 157]
75:25		0.85 (0.78, 0.88)	0.81		0.85 (0.82, 0.88)		0.87 (0.79, 0.98)	0.98	[96,98,107, 108,136, 141,145]
65:35		0.85 (0.79, 0.88)	0.80 (0.79, 0.81)		0.84 (0.81, 0.88)	0.80	0.83 (0.79, 0.86)		[96,98,103, 107,108, 110,136, 145]
50:50	0.79	0.84 (0.77, 0.88)	0.78 (0.75, 0.81)	0.78	0.80 (0.65, 0.87)	0.78 (0.66, 0.85)	0.82 (0.76, 0.98)	0.86 (0.77, 0.95)	[96,98,103, 107,108, 110,117, 136,141, 145,157]

Table 7

The AUC performance for different SE methods with different supervised ML algorithms on imbalanced CMS Part B datasets.

Method	CatB (min, max)	GBT (min, max)	LightGBT (min, max)	LogR (min, max)	NN (min, max)	RF (min, max)	XGB (min, max)	Refe-rence
None	0.80	0.84 (0.82, 0.85)	0.77 (0.71, 0.83)	0.75	0.78 (0.73, 0.83)	0.79 (0.77, 0.82)	0.84	[116,128, 139]
CBOW		0.87						[111]
GloVe		0.87		0.81	0.85	0.85		[116]
Hcpcs2Vec	0.82	0.85 (0.84, 0.87)	0.84	0.81	0.84 (0.83, 0.85)	0.83 (0.80, 0.86)	0.87	[116,128, 139]
Med-W2V		0.86		0.81	0.85	0.85		[116]
OHE	0.79	0.84 (0.83, 0.85)	0.83	0.81	0.84 (0.83, 0.85)	0.79 (0.77, 0.83)	0.85	[111,116, 128,139]
SG		0.87						[111]

Taha et al. [176]. These techniques use statistical attributes, predefined ML algorithms, or a combination of both, to determine the optimal subsets of features that improve the classification accuracy or clustering efficacy.

The ongoing research into feature selection methodologies and SEs within the realm of fraud detection from healthcare claims highlights the opportunity to improve the effectiveness of ML algorithms in this field.

5.5. Machine learning approaches

ML refers to the application of algorithms to data to extract patterns and generate insights through mathematical and statistical analysis. It primarily aids decision-making by unearthing valuable information hidden within data. There are three main categories of ML models: supervised, semi-supervised and unsupervised.

In supervised learning, algorithms are trained on labeled data, using input-output pairs to learn a mapping between inputs and the desired outputs. The objective is to use the learned mapping to make predictions on new and unseen data. Two major applications of supervised

learning are classification (predicting categorical values or classes) and regression (predicting numerical, continuous or real values). In semi-supervised learning, on the other hand, only a portion of the data is labeled for learning, while in unsupervised learning no labels are available to assist in the learning process. A few key applications of unsupervised learning are clustering (grouping data together), outlier detection (discovering anomalies) and association rule mining (ARM).

From the reviewed papers, 41 studies employed unsupervised, 94 studies used supervised, and 12 studies used unique hybrid approaches. Note that the number of studies (147) is more than the number of reviewed papers (137) since seven papers employed both unsupervised and supervised approaches to compare model performances on the same datasets [8,53,74,89,104,113,122], while He et al. [19] used two datasets for two classification applications and Chandola et al. [13] used three different datasets for two clustering and one classification application.

5.5.1. Supervised learning

Supervised learning techniques are preferred when labels are available, as the presence of these labels augment the learning process.

Table 8
The distribution of supervised algorithms.

Algo/Year	2000–04	2005–09	2010–14	2015–19	2020–	Total
DT		3		18	16	37
GBT				4	29	33
kNN	1			1	6	8
LogR		1	1	6	29	37
NB				6	6	12
NN		2		7	30	39
RF				9	36	45
SVC				4	12	16

By leveraging these labels, algorithms can more appropriately discern patterns within distinct classes. Two major applications of supervised learning are regression and classification models. While regression has been applied within the context of hybrid models in the research papers referenced earlier, this section focuses exclusively on classification models.

Classification Algorithms:

Classification algorithms learn from labeled data to predict a categorical outcome, essentially determining the group to which a particular instance belongs. In health insurance fraud detection domain, these algorithms are predominantly used to predict if a provider has engaged in fraudulent activities. In the US, the Office of Inspector General's List of Excluded Individuals and Entities (LEIE) are typically used to label fraudulent providers in the CMS data.

Popular algorithms used in classification include Support vector classifier (SVC), logistic regression (LogR), Random forest (RF), decision trees (DTs), and gradient boosted trees (GBTs) or models (GBMs). Variants of GBTs includes extreme gradient boosting (XGB), CatBoost (CatB) and LightGBM [62,117,128,188]. Refer to Table 9 for detailed references. Other classification algorithms include k-nearest neighbor (kNN) [19,98,127,143,149,151,158,176], Naïve Bayes (NB) [74,98,104,121,127,137,149,165] and its variants multinomial Naïve Bayes (MNB) [102] and Gaussian Naïve Bayes (GNB) [118,151,167], neural networks (NNs) [51,62,63,94,101,103,118,120,135,138] and its variant deep neural networks (DNNs) [65,100,104,110], DAC [74,127], LDA [151,158], QDA [151]. More recent developments of DL algorithms that have been used in this domain include Long Short-Term Memory (LSTMs) [132,173], Convolutional Neural Network (CNNs) [123,147,149,173], Simple Word Embeddings-based Models (SWEM), Gated Recurrent Units (GRUs), and Transformers [173]. Autoencoders (AE) have been used in hybrid settings, where they are used together with NNs [104,146] or tree based algorithms [150].

Table 8 depicts the frequency of supervised algorithms' usage binned into 5-year intervals. As anticipated, there is an upward trend in usage over time, reflecting the growing popularity of ML. Notably, tree-based algorithms like RFs and GBTs emerge as the most frequently used, followed by NNs and LogR. This preference may suggest that these models adequately represent the inherent feature of the available data. Additionally, NNs are gaining in popularity, which illustrates the drive towards using more advanced DL algorithms in an effort to enhance fraud detection from healthcare claims, including CNN, graph NN (GNNs) [152,153,161,163] and graph convolutional NN (GCNN) [54,65,89,147]. As NNs and DL algorithms are very diverse, the abbreviation NN is used as an umbrella term in the tables in this paper.

Table 9 provide the references to the most frequently used algorithms as identified in Table 8.

Significance Testing:

The statistical significance of the model performance is evaluated by using statistical tests. Only twelve of the 137 articles use significant testing, of which one study uses Fisher–Yates in an unsupervised setting [66] and the remainder in an supervised learning setting. Mainly both analysis of variance (ANOVA) and post hoc Tukey's honestly

Table 9
The distribution of frequently used supervised algorithms.

Superv. Algo	# Appli	Reference
DT	37	[61,62,93–98,100,101,103,104,106,107,111,115–118,120,121,127,134,136–138,138,145,151,154,156,159,161,164,165,167,168]
GBT	33	[53,62,64,95,104,106–109,111,114,116,117,119,128,131,139–142,144,146,150,151,154,157,159,161,164,166,173–175]
LogR	37	[63,74,95,97,98,101,106–108,116–119,130,133,136,137,144–146,149,151,153,154,156,157,159,160,162–165,167–170,173]
NN	35	[51,54,62,63,65,89,94,100,101,103,104,110,115,116,118–120,123,126,127,135,138,138,139,146,147,149,152,153,157,158,161,163,166,173]
RF	45	[62–65,74,95,96,98,104,106–108,116,117,119,127,128,130,131,133,134,138,138–141,143,144,146,148,149,151,153,154,156,157,160–164,167–170]
SVC	16	[74,115,121,127,136–138,138,143,145,151,155,160,162,170,176]

significant difference tests (THSDT) have been used together [94–96,103,104,106,107,117,120], and confidence intervals (CIs) [111] and THSDT [116] have each been used in isolation. While ANOVA determines whether the means of several features are equal, THSDT determines which feature means significantly differ from others [95]. As 9% of the articles employ significance testing, further details are not reported.

Performance Evaluation with Supervised ML Models:

Performance evaluation is crucial in ML model development to gauge the relative effectiveness of various trained models. A wide variety of evaluation metrics are used in literature. Popular measurements for classification applications include the area under the receiver operating characteristic curve (AUC or ROC AUC), area under precision–recall curve (PR AUC), F-measure or F1-score, and accuracy or agreement rate. Other measurements used for performance evaluation are G-measure, true positive rate (TPR, also called “recall” or “sensitivity”), true negative rate (TNR, also called “specificity”), false positive rate (FPR, also called “fall-out”), false negative rate (FNR), positive predictive value (PPV, also called “precision” or “hit-rate”), and negative predictive value (NPV) [18,62,70,94,110,122,123]. In recent works, the PR AUC is favored for performance evaluations since studies have indicated that the PR AUC metric is more informative than the ROC AUC metric when comparing the classification performance of multiple classifiers on imbalanced datasets [130], however only one unsupervised [59] and four supervised [130,169,173,175] studies reported on PR AUC results, hence it is not sufficient to allow for comparison and was omitted in the tables.

Table 10 illustrates the top performing algorithms on CMS Part B data across various SEs, different sampling methods and several sampling distributions. According to Table 6, when evaluating algorithm performance for classifying fraudulent providers on the imbalanced CMS Part B data, XGBs and RFs are the best performers with an average AUC of 0.83 and 0.83 respectively whilst for balanced CMS part B data, XGBs and RFs are the best performers again with an average AUC of 0.98 each. From the literature it is clear that higher algorithm performance is achieved on balanced datasets.

Based on the extensive analysis by Bauder and Khoshgoftaar [107] as well as looking at Tables 5 and 10, RUS emerges as the top-performing sampling method for NNs, LogR and tree-based models across all sampling ratios. It is interesting to note that SEs are not amongst the top performing models.

Table 10
The performance of the best ML algorithms on CMS part B data.

Algorithm	SE	Ratio	AUC	References
XGB	No	95:5 RUS	0.98	[141]
RF	No	50:50 RUS	0.98	[141]
CatB	No	50:50 RUS	0.91	[114]
DT	No	80:20 RUS	0.88	[145]
LogR	No	50:50 RUS	0.87	[157]
NN	No	50:50 RUS	0.86	[157]

Validation of Supervised ML Models:

Two types of validation are typically used with supervised learning: the use of a test set, and k-fold cross validation. Sometimes a combination of both are employed. In the first case the original dataset is divided into a training, validation and test sets. The model is trained on the training set and its hyper parameters are tuned for optimal performance on the validation set. Once the model exhibits satisfactory performance on the validation set, its effectiveness is further assessed using the test set. If only two subsets are presented by the literature, they typically represent the training and test sets. In k-fold cross validation (k FCV), the dataset is divided into k folds where $k - 1$ folds are used for training and the k th fold is used for testing the model. This process is repeated k times, using each fold as the test set once, and the average performance of the model is subsequently calculated and reported [12]. A variation, stratified kFCV (k FSCV), ensures that each of the k folds retains a class distribution that mirrors the original dataset's class balance ratio [95]. From the 95 supervised learning studies, 50% employed 5 FCV and 35% used the test/validation set to determine model performance.

5.5.2. Unsupervised learning

Unsupervised learning techniques are especially favored in health insurance fraud detection due to the frequent lack of labeled data in many real-world applications. These techniques excel at discerning patterns of anomalies and outliers from unlabeled data. In this section we discuss both outlier detection (OD) and statistical analysis (SA) techniques, which are often used simultaneously.

Outlier Detection Algorithms:

OD refers to the identification of rare instances or observations which deviate significantly from the majority of the data. Clustering algorithms are often utilized to detect outliers since they aim to group similar instances into groups called clusters, and indicate outliers - the instances that do not cluster with other data points. As an exception, 2 of the 41 unsupervised studies employed ARM [66,69], and 4 distinct studies compared performance of both supervised and unsupervised approaches [53,74,100,104], while the rest relied on clustering algorithms. While a wide range of standard OD algorithms were applied in the reviewed literature, there are also instances where custom algorithms (CA) employing both OD and SA techniques were used. Interestingly half of the studies using OD algorithms also incorporated SA to support their outlier analysis.

Popular OD algorithms include k-means clustering (KMC) [15,58,66,74,76,83,100], local outlier factor (LOF) [12,53,58,60,68,72,75,81,83,85,104,143], PCA [74,80], one class support vector machine (O-SVM) [67,81,85], self organizing maps (SOMs), hierarchical clustering analysis or analytic hierarchical processing (AHP) [50,80,86], expectation maximization (EM) together with Gaussian mixture models (GMMs) [73,160], Mahalanobis distance (MD) [52,104], unsupervised k-nearest neighbors (kNN) [55,72,74,75,81,104], unsupervised random forest (RF) - specifically, the isolation forest (IF) [53,58,59,74,75,81,83,90,92,143], repeated bisections, repeated bisections with global optimization, Bayesian co-clustering, direct k-way clustering, logistic binomial regression (LBR), and density based clustering techniques such as density-based spatial clustering of applications with noise (DBSCAN) [58,84,88] along with its three variants - OPTICS, DENCLUE and GDBSCAN [84]. Additional techniques include non-negative

Table 11
The distribution of unsupervised algorithms.

Algo/Year	2000–04	2005–09	2010–14	2015–19	2020–	Total
GMM	1			3	1	5
IF				2	8	10
KMC			1	3	3	7
kNN			2	3	1	6
LOF		1	1	3	7	12
NN				3	8	11

matrix factorization [57], ARM [66,69], as well as Latent Dirichlet Allocation (LDA) for text mining and graphing algorithms for network analysis [13].

Table 11 illustrates the frequency of utilization for unsupervised algorithms, binned into 5-year intervals. As expected, there is an uptick in usage over time, mirroring the popularity gain of ML owing to its use in applications such as ours. It is interesting to note that LOF [12,53,58,68,72,75,81,83,85,104,143] and IF [53,58,59,74,75,81,83,90,92] emerge as the most used algorithms.

Additionally, the increasing usage of autoencoders (AEs) [59,75,88,90,104,143], unsupervised NNs [85,87] and other DL algorithms including LSTM [59,85] and GNN [89] underscore the ongoing shift towards leveraging more advanced ML algorithms to improve fraud detection efficacy.

Statistical Methods:

In this study, SA is included under the umbrella of unsupervised learning because in our review of research papers we observed that it is often paired with OD to enhance the detection of outliers in the data. In this context, SA refers to a wide array of general statistical techniques including, but not limited to: mean, median, mean or median absolute deviation, mode quantiles, inter-quantile ranges, maximum value, minimum value, z-scores, Gaussian distributions, and standard deviations. More specialized SA techniques and algorithms include Fisher–Yates significance testing, GCross algorithm, Kaplan–Meier edge corrections, Spearman rank-order correlation [66], box-plots [8], hidden Markov models (HMMs) [55,125], Local Correlation Integral (LOCI) [55], fully Bayesian probability models, distance based measures, density based measures, Chi-square statistics, Fisher's Exact Test, Polymorphous (M-of-N) logic [71], Wilcoxon Mann–Whitney test [72], outlier scores (Weighted rank, binomial and standardized residuals), Weiszfeld algorithm, Grubbs test [12], t-test, entropy-reduction, and discriminant analysis (DA) [97].

Validation of Unsupervised ML Models:

One of the primary challenges of employing unsupervised learning is the difficulty in evaluating and validating model performance. This is due to the lack of availability of labeled datasets that can be used for validation and test. To circumvent this, two predominant methods have been adopted: annotations provided by subject matter experts (SMEs) [8,50,70], or the use of any available labeled data solely for model evaluation (and not used for model training) [59,71]. From the 42 unsupervised learning studies, 20% employed SMEs and 10% used a validation dataset to determine model performance.

5.5.3. Hybrid learning

In this paper, the term 'hybrid learning' refers to an ML approach that sequentially employs supervised learning followed by unsupervised learning to determine the target variable - specifically, data outliers that indicate potential fraudulent behavior. This approach is distinct from semi-supervised learning since the supervised learning component learns and predicts a variable that is indirectly related to fraud. Subsequently, unsupervised learning or outlier detection methods identify outliers from this supervised model, which are then flagged for potential fraudulent behavior investigations.

Hybrid Regression Models:

Eight papers utilized regression models followed by OD (with or without using SA) [8,10–12,14,16,87,91]. The regression algorithms

include linear regression (LR), Multivariate Adaptive Regression Splines (MARS), support vector regression (SVR), and Generalized Linear Model (GLM). A typical approach involves fitting a regression to the frequency or monetary value of reimbursed claims over time, where outliers are identified as potential fraud [8]. Other parameters, such as the reimbursed amounts for claims, the number of services performed per service type, and the count of distinct Medicare beneficiary/per day services aligned against different providers, have also been considered [10,12]. Bauder and Khoshgoftaar [10] compared the performance of MARS, SVR, GLM and two variants of GLM – GLMNET and BayesGLM – in predicting the expected payment amount for claims to determine possible deviations.

Various techniques have been adopted for outlier detection. Van Capelleveen et al. [8,16] used a standard deviation exceeding 2.33 to identify outliers from a LR model, and identified anomalies lying outside the inter-quartile ranges in boxplots. In other studies, residuals (difference between the predicted and actual values) coupled with the residuals' standard deviation are used to create the thresholds, which are then used to detect possible fraudulent activities [10]. After model fitting, MARS yields residuals, and Bauder and Khoshgoftaar [11,12] generated studentized residuals as input for an outlier detection model. The studentized residuals are computed by applying the Student's *t*-statistic and then normalizing residual outputs to facilitate a direct comparison.

Ko et al. [14] utilized a LR model to categorize instances where the actual claim payment exceeded the predicted claim payment amount. These instances were then divided into quartiles. By contrasting the highest with the lowest quartiles, they aimed to identify potential outliers.

Common metrics for assessing regression algorithms encompass Pearson's correlation coefficient (*r*), coefficient of determination (R^2), root mean square error (RMSE), mean absolute error (MAE) and mean absolute percentage error (MAPE) [10–12]. For hybrid applications, performance is typically evaluated using the F1-score and accuracy, given the primary objective of accurately identifying fraudulent instances. As the hybrid regression studies have vastly different reported metrics, readers are encouraged to consult the references for additional information.

Hybrid Classification Models:

Four reviewed papers applied hybrid classification models, where the classification model learns labels other than explicit fraudulent labels and OD is subsequently used to determine potential fraudsters [102, 105, 123, 172]. Two of these applications aim to predict the provider's specialty and if there is a mismatch between the predicted and actual specialty (outliers), that provider is flagged as potentially fraudulent [102, 105]. These studies employed the Multinomial Naïve Bayes (MNB) algorithm for classification. Zhang et al. [123] leveraged a text mining application to determine the appropriate treatment a patient should have received and then compared it to the actual treatment to identify outliers in provider treatment patterns. Ghani and Kumar [172] used a classifier as initial data partition and subsequently employed OD methods for more finer-grained partitioning, aiming to discern potential fraudulent instances within payment data.

5.6. The challenges of fraud detection from healthcare claims

Detecting health insurance fraud is a notoriously difficult task due to the intricate nature of the field, the wide variety of transactional data formats, and the hidden interactions between participants, such as those found in fraud rings.

5.6.1. Data quality and consistency challenges

ML-based fraud detection systems are inherently reliant on access to high-quality and detailed data, allowing for feature representations that mirror the domain's complexity. The following list highlights potential inadequacies in the data available for fraud detection [2]:

1. **Data Inconsistencies:** The complexities and variety of procedure and service codes can lead to instances where a single claim is documented under diverse codes.
2. **Data Formatting:** Healthcare providers employ unique systems for data generation and storage, each with its distinct format, attributes, and semantics. This necessitates system-specific knowledge to audit or use claim data from these systems. Additionally, this leads to the lack of standardized data repositories.
3. **Data Integration:** In our review, we observe that research has often been performed on individual data sources, from either public data repositories or private health insurance companies, or newly created datasets based on integration of features from related data repositories. Although combined datasets might offer valuable information absent from singular data sources, they are privately created and managed, so standardized, centralized and public versions are nonexistent.
4. **Privacy Concerns:** Owing to privacy and security concerns, healthcare entities (providers, insurers, etc.) and patients are often reluctant to share data. This diminishes the quality and quantity of available healthcare claims data and leads to potential loss of critical information.
5. **Scarcity of Fraudulent Cases:** Due to the sheer volume of healthcare transactions and the complexity of the healthcare system, the process of auditing these transactions and correctly identifying fraudulent activities is time-consuming and delivers few fraudulent cases compared to the overwhelming volume of transactions. Consequently, there is a notable scarcity of known fraudulent cases for ML systems to learn from and predict future fraudulent cases.

The diverse range of healthcare actors and their behaviors, along with the complexities of databases and healthcare transactions (such as procedure and service codes, drug prescriptions, medical equipment orders, etc.), present a challenging environment for AI to navigate. Regarding the CMS datasets, the PUP_POS dataset comprises roughly 75 features with an average of 1.3 million instances of providers annually [180], while the PUP_D includes about 85 features and an average of 1 million provider instances per year [181]. Additionally, the DMEPOS dataset contains approximately 98 features and an average of 300,000 provider instances annually [182]. Moreover, providers are connected across datasets using unique identification numbers, a factor that must be considered during data integration. This highlights both the extensive size of the datasets to be processed and the challenge of choosing the most informative features to aid in using ML to detect fraudulent behavior. Consequently, this situation presents familiar ML challenges, including the curse of dimensionality, lengthy training times, and increased computational and memory requirements.

While some of these challenges and their possible solutions (e.g. federated learning to address privacy concerns) are active areas of research, our review primarily highlights strategies to address the rarity of fraudulent cases, especially when training supervised ML models. We delve deeper into these strategies in the next section.

5.6.2. Machine learning standardization

The dataset sources exhibit considerable variability in their composition and preprocessing methods prior to the application of ML algorithms, as detailed in Sections 5.2, 5.3, 5.4, and Table 3. Additionally, the range of available ML algorithms, discussed in Section 5.5, further complicates the ability to uniformly compare performances across datasets due to the lack of a standardized approach. In the realm of healthcare claims, fraud detection methods are still evolving. Recent studies, for instance, have been experimenting with the novel application of SEs and feature selection techniques to identify the most relevant dataset features. These efforts aim to mitigate the challenge of high dimensionality and enhance fraud detection efficiency. However, this field is still in its nascent stages, and best practices for ML-based fraud detection on real-world data are yet to be established.

5.6.3. The scarcity of known fraudulent cases

In the US, the Government Accountability Office (GAO) stipulates guidelines and recommendations for auditing healthcare claims and ensuring data privacy. The OIG independently audits and reviews GAO programs, in addition to investigating allegations of fraud, waste, mismanagement, and legal violations to uphold the agency's efficacy [189]. The auditing process typically involves a subject matter expert (SME) re-coding an original claim and then comparing it to the provider's initial submission. Any discrepancies found may prompt further investigations into potential fraud [2]. The volume of annual healthcare claims is staggering and continues to grow exponentially. In 2021, for instance, the GAO reported 61.6 million Medicare beneficiaries in the US, resulting in millions of claims each year [190]. Given the immense volume of annual claims, the available audit data for fraud detection is often limited and potentially outdated, as auditing is a manual and time-intensive task.

For data sourced from private entities or private health insurance companies, labels akin to those in the OIG's LEIE list might not be readily available, or their acquisition could be subject to delays. In these cases, it becomes necessary to engage SMEs proficient in auditing and labeling fraudulent cases. However, relying on SMEs presents its own challenges. Not only are SMEs costly to employ, but finding those who possess both in-depth knowledge and experience of the system and its processes, as well as a willingness to dedicate themselves to the auditing and labeling tasks, can be a difficult and complex endeavor.

In the context of ML, the scarcity of known fraudulent cases contributes to class imbalance issues. While negative or non-fraudulent cases are available in abundance, the pool of positive or fraudulent cases — which is essential for learning the characteristics of fraudulent transactions in predictive systems — is very limited. This imbalance presents a significant hurdle in fraud detection. In our literature review, we discovered that unlabeled datasets typically contain around six million instances, while labeled datasets average seven million instances. The ratio of non-fraudulent to fraudulent cases in these datasets is commonly 99.5:0.5. This pattern holds true for datasets specific to the US. However, for non-US datasets, we found a notable difference: unlabeled datasets also average six million instances, but labeled non-US datasets are smaller, averaging around 200,000 instances. Moreover, the ratio of non-fraudulent to fraudulent cases in these non-US datasets shifts to 88:12. We found that there is not necessarily a correlation between dataset size and ratios, as a ratio of 99.99:0.01 is observed for dataset sizes ranging from 1 million [134] to 174 million instances [140].

6. Discussions and future considerations

With the various challenges experienced in employing ML for fraud detection from healthcare claims, various opportunities arise for innovative solutions to these challenges. Bauder and Khoshgoftaar [31] specifically focus on Medicare data and propose several recommendations to address privacy concerns and manage missing data in datasets, thereby enabling more effective fraud detection. While certain algorithms, such as XGBoost and Naive Bayes are capable of handling missing data, other traditional approaches such as logistic regression or SVMs are not equipped for this. Although missing data can sometimes be imputed from existing values, these situations require careful evaluation on a case-by-case basis. In another study by Bauder et al. [2], various methods for label generation are suggested, including text mining and network or graph analysis. The authors advocate for a deeper understanding of provider interactions and improved integration of data sources. Their approach aims to uncover subtle patterns and advance the effectiveness of fraud detection in healthcare claims.

Kumaraswamy et al. [41] identify four key areas of improvement within this domain. First, they advocate for a unified framework amalgamating various research methods for fraud detection (across data sources, application areas, detection methods and performance metrics), aiding investigators in their work. Secondly, models should adapt

to the diverse feature sets present in regional, national, or private databases, ensuring consistent performance even with missing data. Thirdly, they stress the importance of standardizing data aggregation and preprocessing steps. Lastly, they emphasize the need for researchers to be informed about the phases executed after the detection of potential fraud to assist in the continuous improvement of the fraud detection models (online learning).

In other areas of fraud detection such as credit card fraud, deep reinforcement learning (RL) [191,192] and adversarial learning [193–196] have been shown to offer innovative solutions when fraud labels are scarce, datasets are imbalanced, and fraudsters are improving their strategies over time. Such techniques could potentially be applied to healthcare claims fraud detection. However, there is a notable contrast with credit card fraud, which typically occurs in real-time at points of sale. Healthcare claims fraud, on the other hand, tends to involve more premeditated actions with schemes spanning longer time spans.

In adversarial models both an adversary (fraudster) and classifier (fraud detector) have to be trained. These models can vary in complexity across several levels:

1. The classifier implements an advanced selection strategy using existing data, which facilitates its iterative training and improvement.
2. The classifier is continuously refined. The main challenges in this stage are determining the appropriate timings and methodologies for improving the classifier.
3. Both the adversary and classifier evolve over time to enhance their capabilities. Key challenges at this level include realistically modeling these evolving behaviors, identifying potential biases in the process, and assessing the reliability of the modeled adversary behavior.

The literature since 2020 includes the use of generative adversarial networks (GANs) in fraud detection from healthcare claims. Naidoo and Marivate [113] addressed the lack of fraudulent labels by training a GAN to cluster fraudulent data based on an anomaly score which is then used to train four supervised learning models to enhance fraud classification performance. Nugraha et al. [197] used a Conditional Tabular GAN (CTGAN) as an oversampling method to generate additional data for the minority (fraudulent) classes to balance the data before training 17 classifiers on the data. These applications can function as stepping stones for using GANs and RL to address complexities within healthcare insurance fraud detection. A major drawback of using GANs is the challenge of modeling fraudster behavior realistically, which is related to the challenge of overcoming privacy concerns to acquire data containing indicators of fraudster behavior.

Another approach that could benefit our understanding of the behavior of fraudsters in a heterogeneous relationship space of multiple actors is the use of graphs and graph neural networks (GNNs) [198–200]. Applications first surfaced in the late 2010s [54,65,89,152,153,161,163], and include papers in which GNNs are used as part of the data pre-processing activities or for fraud prediction purposes. GNNs have been gaining popularity within literature due to their ability to use graph networks to capture underlying interactions in data. GNNs could be particularly useful in fraud detection from healthcare claims since it enables the creation of rich datasets instead of sparse matrices where categorical features are concerned. Furthermore, GNNs have the additional benefit of potentially capturing conspiracy fraud and fraud rings between the various actors in the insurance claims system.

Other advancement in terms of ML applications could include enhancing the explainability of ML algorithms. This aligns with the principles outlined in the recently established AI Bill of Rights in the US [201]. The bill of rights is founded on five key principles, namely safe and effective systems; algorithmic discrimination protections; data privacy; notice and explanation; and human alternatives, consideration, and fallback options. These principles are designed to guide AI and

ML developers in the creation, usage, and deployment of automated systems, ensuring the development of safe, reliable, and trustworthy models. Regarding explainability, there is an increasing demand to transform ML algorithms from 'black-box' solutions, which are opaque and difficult to interpret, to 'white-box' solutions that are transparent and understandable to humans. This shift is crucial for identifying potential harmful biases and discrimination in models, such as learning incorrect associations for decision-making. Additionally, it can reveal areas for improving algorithm performance, such as highlighting the absence of certain data features [202]. Various techniques, software tools and programming libraries are available to aid in model explainability. Examples include Shapley values [203], LIME values [204], and integrated gradients [205], each offering different approaches to interpret and understand model decisions.

7. Limitations

A potential limitation of this survey is incomplete retrieval or identification of relevant studies that meet the search criteria. As a wide variety of ML methods are employed to conduct ML-based fraud detection from healthcare claims, it is challenging to conduct an inclusive search strategy with an automatic query by keyword searching. This limitation was mitigated by applying the snowballing strategy to the included research.

It is challenging to distinguish the included studies from various works employing statistical methods for fraud detection, since ML and statistical methods are overlapping fields of applications. To address this all the collected papers underwent a full-text review to verify that all the included papers applied recognized ML algorithms or were indicated as overlapping or unique applications. In addition, only fully published, full-text works were evaluated.

Another potential limitation of this survey is its inability to address all technical questions comprehensively. Prior research has evaluated various factors, including feature selection, dataset sampling and balancing, model development and evaluation, and various learning and optimization approaches, all aimed at enhancing the detection of insurance fraud and predictive performance. However, the most effective combination of these factors for detecting fraud in healthcare claims and advancing model development remains unclear. A potential solution for this challenge is to conduct a comprehensive experiment to investigate and evaluate different combinations of these factors using a benchmark dataset or a common, publicly available dataset. Such an experiment should be conducted with full transparency in its methodologies, enabling reliable and clear comparisons between different methods.

8. Conclusions

ML applications have led to a wide variety of innovations in the process of detecting fraud in healthcare claims data. Since the developed models benefit largely from ML capabilities to address the challenges of healthcare claims fraud data, ML methods are a promising route to develop powerful and robust applications. It was concluded that healthcare providers are the most common actor of fraud. Various aspects of ML model development have been considered in fraud detection from healthcare claims literature, including feature selection, sampling methods, dataset balance ratios, model selection, learning approaches and model performance evaluation. It is noted that, although literature was collected from around the globe, the majority of the literature work originates from the US. Various datasets have been used for healthcare fraud detection, however it is noted that CMS datasets from the US are most popular. A variety of data sampling methods have been employed in literature and multiple studies conclude that RUS leads to better performance. GBTs were found to be the best supervised ML algorithms regardless of the use of sampling methods or SEs. Unsupervised approaches are still variable and there is a noticeable increase in the use

of AEs and IFs in healthcare fraud detection. A systematic review of this research has been conducted and current research applications as well as challenges and opportunities have been discussed. These challenges include data quality, ML standardization, and the scarcity of known fraudulent cases. Future research directions include the exploration of deep learning, the explainability of ML algorithms and transparency in the ML pipelines. Further research and development of advanced, sophisticated methods are needed to ensure the continuous improvement of representation learning and ML modeling for fraud detection from healthcare claims.

CRedit authorship contribution statement

Anli du Preez: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Sanmitra Bhattacharya:** Writing – review & editing, Writing – original draft, Conceptualization. **Peter Beling:** Writing – review & editing, Writing – original draft, Resources. **Edward Bowen:** Conceptualization.

Declaration of competing interest

ADP declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this survey paper. SB and EB are employees of Deloitte & Touche LLP.

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